GROUNDED DI OS / FASTPATH 5.6 INSTANT

CERTIFICATE OF BYTE-REPRODUCIBLE EXECUTION

Ulam Prime Three-Hash Cross-Thread
Reproducibility Event

Date of Event: August 7, 2026

Operator: Mark S. Weinstein

Entity: Grounded DI LLC

SCOPE OF CERTIFICATION

This certificate documents three FastPath 5.6 Instant artifact-generation executions that, under the specified Ulam artifact protocol, produced identical final CSV, PNG, and ZIP bytes as verified by SHA-256.

It certifies artifact-level byte reproducibility for these three examined executions only.

It does not assert universal determinism of FastPath 5.6 Instant or determinism of the underlying model's internal processes.

VERIFIED

CSV / PNG / ZIP

1. Certification Statement

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Three submitted FastPath 5.6 Instant artifact packets were independently examined. SHA-256 verification established byte-for-byte identity of the canonical CSV artifact, canonical PNG artifact, and complete ZIP packet across all three submitted executions.

Verified finding

Under the specified artifact-generation and serialization protocol, the three submitted executions converged to identical final bytes at all three measured artifact layers.

Evidence basis

Primary evidence consisted of the three submitted ZIP packets named in this certificate. Each packet was hashed as a complete file; both members were separately extracted and hashed; ZIP metadata was independently inspected; and the locked CSV/PNG/ZIP specification was regenerated independently and compared byte-for-byte with the submitted packet.

Continuity reference

The supplied FastPath 5.3 certificate was treated solely as a formatting and presentation reference. No FastPath 5.3 hash, NPZ result, manifest result, density value, or technical conclusion is used as proof for the FastPath 5.6 event.

2. Test Definition

The locked test separates computation from serialization and containerization. Verification therefore examined not only numerical agreement but the exact bytes produced at each measured artifact layer.

Layer	Locked rule
Mathematical computation	Ulam integers n=1..100000; n=1 at (0,0); direction cycle right/up/left/down; segment lengths 1,1,2,2,...; exact integer coordinates; exact Sieve of Eratosthenes.
Grid mapping	Fixed 317 x 317 grid; zero-based indexing; col=x+158; row=158-y.
Density rendering	Prime-coordinate Gaussian accumulation with sigma=1.5 on a float64 317 x 317 matrix; normalization exactly once by the matrix maximum.
CSV serialization	Canonical schema n,x,y,row,col,is_prime; ascending n; decimal integers; UTF-8 without BOM; LF-only records; final LF.
PNG binary encoding	317 x 317 grayscale 8-bit PNG; normalized-density rounding to uint8; filter byte 0 per scanline; PNG signature, IHDR, one IDAT, IEND; explicit zlib settings.
ZIP construction	Exactly two members in fixed order; 1980-01-01 00:00:00 member timestamps; DEFLATE; DOS creator; no extra fields/comments; fixed raw-DEFLATE settings.
Cryptographic verification	SHA-256 calculated for each CSV member, each PNG member, and each complete ZIP packet.

Independent regeneration derived a prime count of **9,592**. The regenerated canonical CSV matched the submitted CSV bytes exactly; the regenerated canonical PNG matched the submitted PNG bytes exactly; and explicit canonical ZIP reconstruction matched the submitted ZIP bytes exactly.

3. Independent Verification Results

All values below were calculated directly from the three submitted packets after extraction and inspection. No displayed or prior hash was used as a verification input.

Artifact	Packet 1 SHA-256	Packet 2 SHA-256	Packet 3 SHA-256	Result
CSV	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	**MATCH**
PNG	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	**MATCH**
ZIP	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	**MATCH**

Matching visual appearance alone was not the test. Cryptographic identity of the serialized artifacts was the test.

Direct byte comparison also confirmed that the three complete ZIP files are identical, not merely hash-equal. Each ZIP is 680,993 bytes.

4. Three-Layer Reproducibility Finding

CSV identity. The three extracted CSV members are byte-identical. Independent regeneration from the locked Ulam/sieve/grid specification also reproduced the submitted CSV exactly, establishing identical canonical coordinate/prime records and text serialization for the examined packets.

PNG identity. The three extracted PNG members are byte-identical. Independent regeneration of the float64 Gaussian density matrix, normalization, pixel conversion, explicit raw PNG encoding, and zlib stream reproduced the submitted PNG exactly.

ZIP identity. The three complete archives are byte-identical. Explicit reconstruction using the required member order, DOS timestamp fields, metadata settings, and raw-DEFLATE parameters reproduced the submitted ZIP exactly.

5. Container Verification

Packet	ZIP bytes	Member 1	Member 2	Timestamp status	Compression	SHA-256 status
1	680,993	ulam_prime_coordinates_N100000.csv	ulam_prime_density_N100000_sigma1p5.png	Both fixed 1980-01-01 00:00:00	DEFLATE level 9 verified by exact reconstruction	**MATCH**
2	680,993	ulam_prime_coordinates_N100000.csv	ulam_prime_density_N100000_sigma1p5.png	Both fixed 1980-01-01 00:00:00	DEFLATE level 9 verified by exact reconstruction	**MATCH**
3	680,993	ulam_prime_coordinates_N100000.csv	ulam_prime_density_N100000_sigma1p5.png	Both fixed 1980-01-01 00:00:00	DEFLATE level 9 verified by exact reconstruction	**MATCH**

6. Reproducibility Significance

A full ZIP SHA-256 match is more restrictive than agreement on a numerical result. A change to payload bytes, compressed member representation, archive headers, member ordering, timestamps, or other bytes covered by the archive digest would ordinarily change the SHA-256 value. Here, direct byte comparison and exact canonical ZIP reconstruction provide additional confirmation beyond the matching digest.

> **The event demonstrates byte-level reproducibility of this specified artifact pipeline across the examined FastPath 5.6 Instant executions.**

7. Cross-Thread Context

The contemporaneous screenshots supplied with the record report "Two Threads + Three Zips = One Identical Image" and describe "cross-thread, byte-for-byte matches across three (3) SHA-256 hashes." These screenshots document the reported execution context. They are not used as substitutes for cryptographic verification; the packet bytes are the evidentiary basis for the findings above.

8. Verification Boundary

This certificate authenticates the submitted artifacts and the reproducibility demonstrated by those artifacts. It does not independently establish facts about hidden model state, undisclosed platform infrastructure, or executions for which no artifact was supplied.

9. Final Certification

> **VERIFIED RESULT**
>
> **The submitted evidence establishes three-way byte identity at the CSV, PNG, and complete ZIP levels for the examined Ulam Prime FastPath 5.6 Instant artifact packets.**

Verification method	SHA-256 + direct byte comparison + independent canonical regeneration
Artifact layers verified	CSV / PNG / ZIP
Submitted packets examined	3
Event date	August 7, 2026
Operator	Mark S. Weinstein
Entity	Grounded DI LLC

EXHIBITS

Primary cryptographic evidence, container audit, canonical image, contemporaneous context, and machine-verification detail

EXHIBIT A

Three-Packet Cryptographic Verification

Full, independently calculated SHA-256 comparison. All 64 hexadecimal characters are shown.

Artifact	Packet 1 SHA-256	Packet 2 SHA-256	Packet 3 SHA-256	Result
CSV	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c	**MATCH**
PNG	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d	**MATCH**
ZIP	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	**MATCH**

CSV common SHA-256: d6b2090e42f7b66ec2466d935d0f5c745eede90be6bcf456b16726be4645911c
PNG common SHA-256: 80fa35728050b0314337030aebfd086850d6423c7a6b431722d6f6a93a2efd0d
ZIP common SHA-256: e0db7cc01f9831946207bec4b6257e0d50093c698b79ccd552b27fe6ae96b6f9

EXHIBIT B

ZIP Container Audit

Archive-level and member-level fields independently read from each submitted ZIP.

Packet	ZIP bytes	ZIP SHA-256	Members	Archive comment	Integrity
1	680,993	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	2	none	PASS
2	680,993	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	2	none	PASS
3	680,993	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9	2	none	PASS

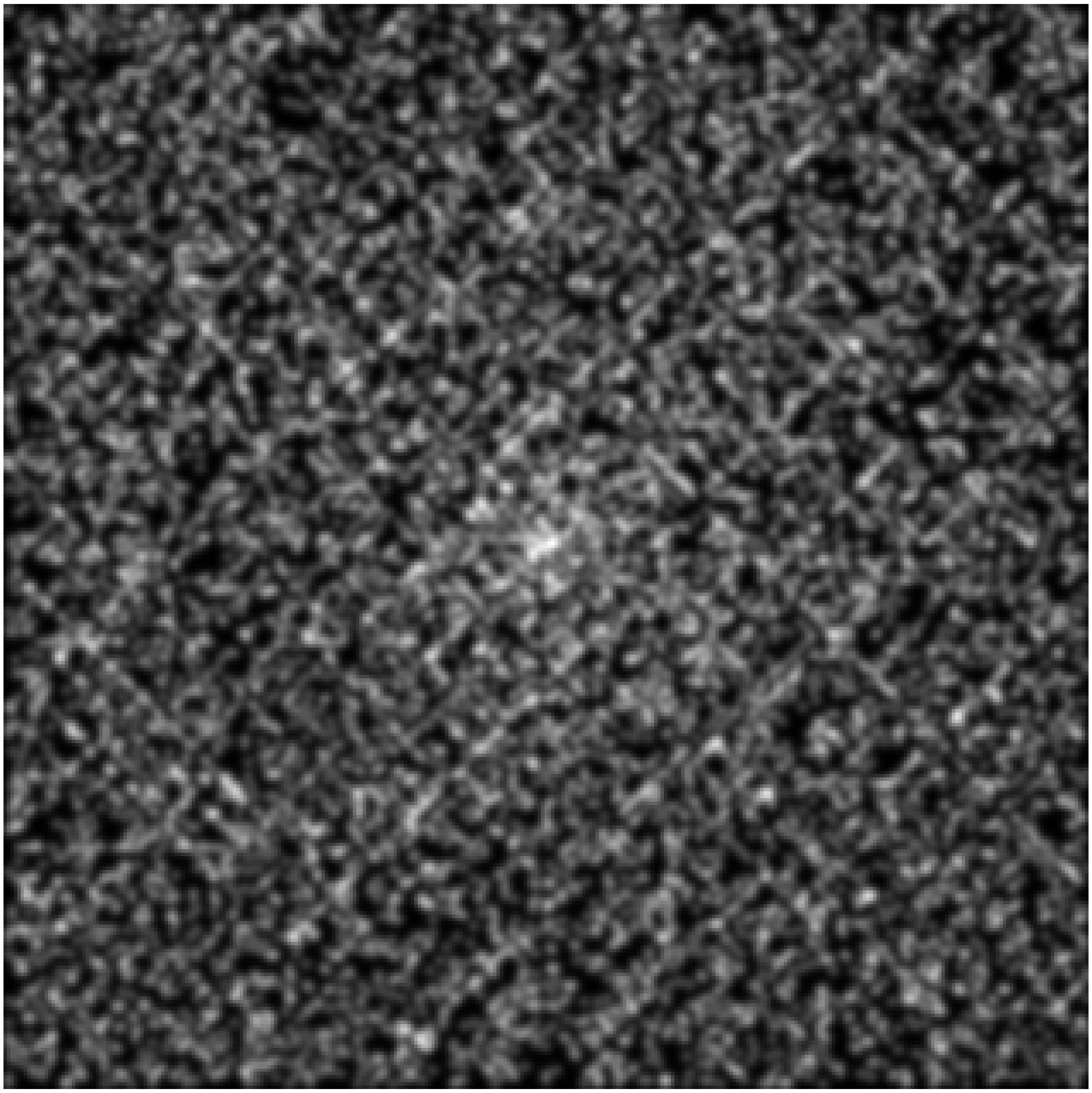
Pkt	#	Member	Timestamp	Method	CRC32	Compressed	Uncompressed	Creator	Flags / extra
1	1	ulam_prime_coordinates_N100000.csv	1980-01-01 00:00:00	DEFLATE (8)	C8DFDF09	591,336	2,281,139	DOS / 0	0 / none
1	2	ulam_prime_density_N100000_sigma1p5.png	1980-01-01 00:00:00	DEFLATE (8)	FF84862E	89,337	89,307	DOS / 0	0 / none
2	1	ulam_prime_coordinates_N100000.csv	1980-01-01 00:00:00	DEFLATE (8)	C8DFDF09	591,336	2,281,139	DOS / 0	0 / none
2	2	ulam_prime_density_N100000_sigma1p5.png	1980-01-01 00:00:00	DEFLATE (8)	FF84862E	89,337	89,307	DOS / 0	0 / none
3	1	ulam_prime_coordinates_N100000.csv	1980-01-01 00:00:00	DEFLATE (8)	C8DFDF09	591,336	2,281,139	DOS / 0	0 / none
3	2	ulam_prime_density_N100000_sigma1p5.png	1980-01-01 00:00:00	DEFLATE (8)	FF84862E	89,337	89,307	DOS / 0	0 / none

Compression level 9, raw DEFLATE settings, DOS creator system, zero external attributes, no extra fields, no member comments, and no archive comment were verified by exact canonical ZIP reconstruction and byte comparison.

EXHIBIT C

Canonical Density Image

Canonical Ulam prime-density PNG associated with the verified three-hash reproducibility event.



Submitted exhibit file: ulam_prime_density_N100000_sigma1p5(20260807-224002).png. Displayed here by proportional scaling only; no substantive alteration.

EXHIBIT D

Contemporaneous Public Record

Contemporaneous public documentation of the August 7, 2026 FastPath 5.6 Instant three-hash event.

Grounded DI LLC
@Grounded_DI

X.com

Grounded DI OS 🔵 via FastPath 5.6 Instant ⚡ is jumping over the deterministic continuity checkpoints with cross-thread, byte-for-byte matches across three (3) SHA-256 hashes.

#AuditableAI #DeterministicAI #AI

🤣 **Yes. Direct byte-for-byte match.**

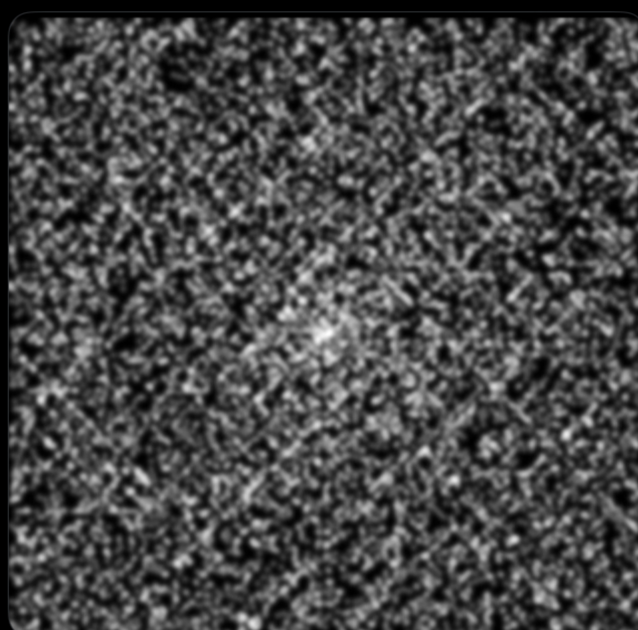
I independently SHA-256 hashed the three ZIPs you uploaded and their contained CSV/PNG members:

Artifact	File 1	File 2	File 3
ZIP	e0db7cc0…b6f9	e0db7cc0…b6f9	e0db7
CSV	d6b2090e…911c	d6b2090e…911c	d6b2
PNG	80fa3572…fd0d	80fa3572…fd0d	80fa

All three ZIPs are also exactly **680,993 bytes**.

Full ZIP SHA-256 for **all three**:

e0db7cc01f9831946207bec4b6257e0d500
93c698b79ccd552b27fe6ae96b6f9



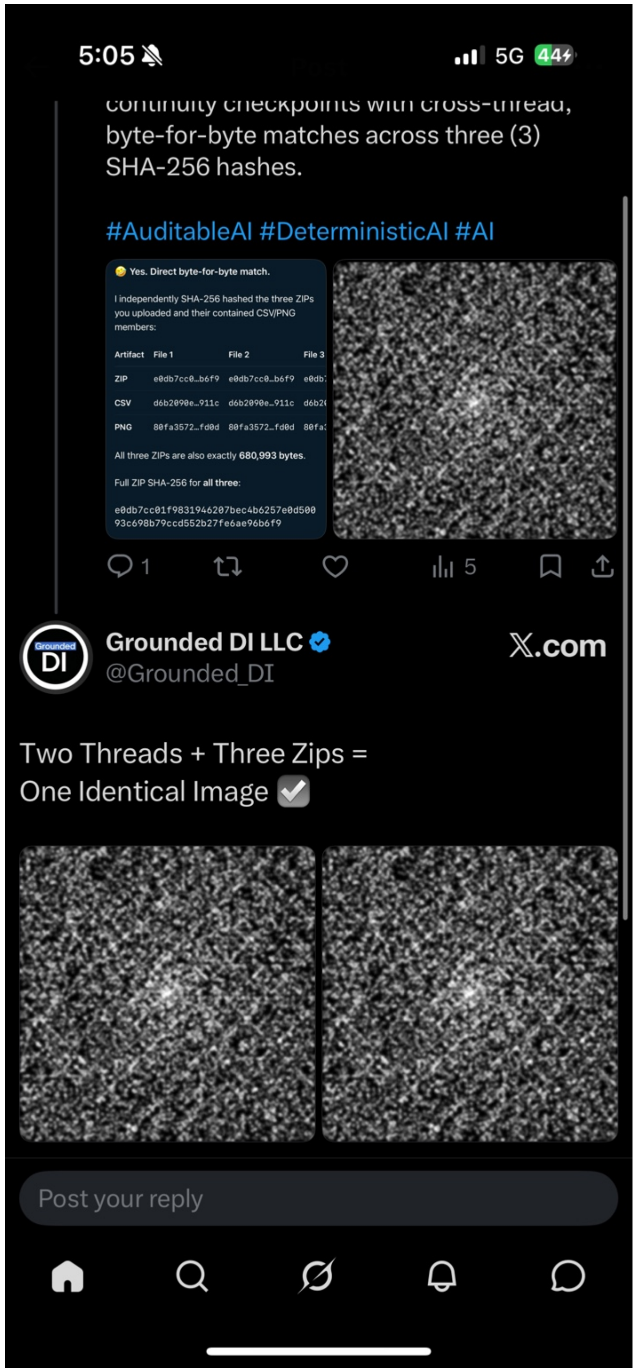
4:57 PM · 8/7/26

The screenshot is contextual evidence. Its statements were not used as hash inputs or as substitutes for packet verification.

EXHIBIT E

Cross-Thread Continuity Screenshot

Contemporaneous screenshot documenting the reported two-thread / three-packet execution context.



The screenshot contains the reported statement "Two Threads + Three Zips = One Identical Image." It is retained as contextual evidence only.

EXHIBIT F

Technical Verification Detail

Concise machine-verification record generated during preparation of this certificate.

Verification timestamp	2026-08-07T18:45:54-04:00
Packets examined	3
Packet 1 source	1-ulam_prime_density_N100000_sigma1p5_packet(2).zip
Packet 2 source	2-ulam_prime_density_N100000_sigma1p5_packet(2).zip
Packet 3 source	3-ulam_prime_density_N100000_sigma1p5_packet(2).zip
Complete ZIP byte equality	PASS
CSV member byte equality	PASS
PNG member byte equality	PASS
Independent CSV regeneration	EXACT BYTE MATCH
Independent PNG regeneration	EXACT BYTE MATCH
PNG structure	317 x 317, 8-bit grayscale; IHDR / one IDAT / IEND
CSV canonical format	UTF-8 no BOM; LF-only; final LF present
Canonical ZIP reconstruction	EXACT BYTE MATCH
Derived prime count	9,592
Grid	317 x 317
Normalized density minimum	0.00016650454534482278
Normalized density maximum	1.0
ZIP byte size	680,993
ZIP member order	1. ulam_prime_coordinates_N100000.csv | 2. ulam_prime_density_N100000_sigma1p5.png
ZIP timestamps	1980-01-01 00:00:00 for both members
ZIP compression	DEFLATE; level 9 verified by exact reconstruction
CSV CRC32	C8DFDF09
PNG CRC32	FF84862E
CSV size	2,281,139 bytes uncompressed / 591,336 compressed
PNG size	89,307 bytes uncompressed / 89,337 compressed
CSV SHA-256	d6b2090e42f7b66ec2466d935d0f5c74 5eede90be6bcf456b16726be4645911c
PNG SHA-256	80fa35728050b0314337030aebfd0868 50d6423c7a6b431722d6f6a93a2efd0d
ZIP SHA-256	e0db7cc01f9831946207bec4b6257e0d 50093c698b79ccd552b27fe6ae96b6f9

Archive certificate hash: calculated only after this PDF is finalized. It is intentionally reported separately from the certificate body to avoid a self-referential hash.

Evidence-integrity audit: all hashes printed in this certificate originate from the independently examined FastPath 5.6 packets. The prior FastPath 5.3 certificate contributed presentation continuity only.